STATISTICS IS THE GRAMMAR OF SCIENCE

PROBABILITY AND STATISTICS

LECTURE – 18

PROBABILITY DISTRIBUTIONS

GAMMA DISTRIBUTION

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GAMMA DISTRIBUTION

The Gamma Distribution was first introduced by Karl Pearson. The **gamma distribution** has many important applications in areas such as reliability theory and it is also used in the analysis of a Poisson process.

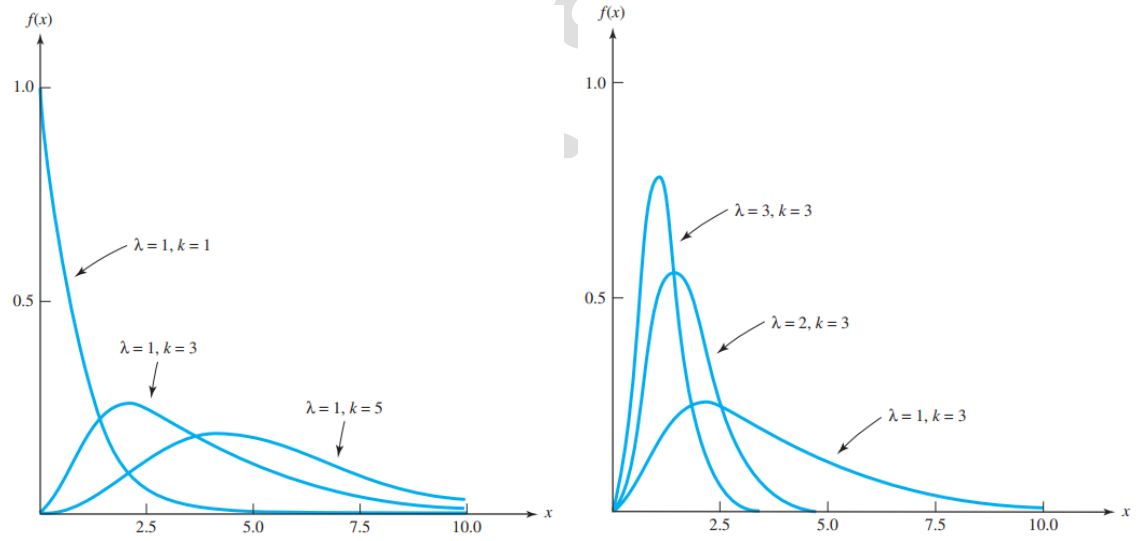
THE GAMMA DISTRIBUTION A continuous random variable X is said to have a gamma distribution with parameters (α,λ)if it has the following probability density function

$$f(x)=\begin{cases}\dfrac{\lambda^{\alpha}}{\Gamma(\alpha)}x^{\alpha-1}e^{-\lambda x} & for\ x\geq o\\ 0 & otherwise\end{cases}$$

$where\ the\ parameter\ \alpha>0\quad and\quad \lambda>0\,.$

The parameter α is often referred to as the *shape* parameter of the gamma distribution, and the parameter λ is referred to as the *scale* parameter of the gamma distribution.

GRAPHS OF SOME GAMMA DISTRIBUTIONS



Note that in the above graphs $\alpha=k$ and the parameter $\lambda=\lambda$.

PROPERTIES OF GAMMA DISTRIBUTION

1- The Mean of Gamma Distribution is $E(X)=\alpha/\lambda$

2- The variance of Gamma Distribution is $V(X)=\alpha/\lambda^{2}$

EXAMPLES OF GAMMA DISTRIBUTION

EXAMPLE-1 Determine the following Gamma Probabilities.

$(a)\ P\left(X \le 3 \middle| \alpha = 2,\ \lambda = 3\right)$

$(b)\ P\left(X \ge 4 \middle| \alpha = 2,\ \lambda = 3\right)$

$(c)\ P\left(3 \le X \le 5 \middle| \alpha = 2,\ \lambda = 3\right)$

SOLUTION Determine the following exponential probabilities.

$(a)\ P\left(X \le 3 \middle| \alpha = 2,\ \lambda = 3\right)$

$$= \int_0^3 \frac{\lambda^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\lambda x} dx = \int_0^3 \frac{(3)^2}{\Gamma(2)} x^{2-1} e^{-3x} dx$$

$$= \frac{9}{1!} \int_0^3 x^{2-1} e^{-3x} dx = 9 \times \int_0^3 x^1 e^{-3x} dx = 0.9987$$

$(b)\ P\left(X \ge 4 \middle| \alpha = 2,\ \lambda = 3\right)$

$$= \int_4^\infty \frac{\lambda^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\lambda x} dx = \int_4^\infty \frac{(3)^2}{\Gamma(2)} x^{2-1} e^{-3x} dx$$

$$= \frac{9}{1!} \int_4^\infty x^{2-1} e^{-3x} dx = 9 \times \int_4^\infty x^1 e^{-3x} dx = 0.00007$$

$(c)\ P\left(3 \le X \le 5 \middle| \alpha = 2,\ \lambda = 3\right)$

$$= \int_3^5 \frac{\lambda^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\lambda x} dx = \int_3^5 \frac{(3)^2}{\Gamma(2)} x^{2-1} e^{-3x} dx$$

$$= \frac{9}{1!} \int_3^5 x^{2-1} e^{-3x} dx = 9 \times \int_3^5 x^1 e^{-3x} dx = 0.0012$$

EXAMPLE-2 Suppose that, on average, the number of β-particles emitted from a radioactive substance is four every second. *If* $\alpha = 2,\ \lambda = 4$ What is the probability that it takes at least 2 seconds before the next two β-particles are emitted?

SOLUTION Let X be the time between now and when the second β-particle is emitted

Here $\alpha = 2,\ \lambda = 4$. *we have to find* $P(X \ge 2) = ?$

$$P(X \ge 2) = \int_2^\infty \frac{\lambda^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\lambda x} dx$$

$$= \int_2^\infty \frac{4^2}{\Gamma(2)} x^{2-1} e^{-4x} dx$$

$$= 16 \int_2^\infty x^1 e^{-4x} dx = 0.003$$

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IMPORTANT CONCEPTS AND FUNCTIONS IN PROBABILITY

GAMMA FUNCTION

$$\Gamma(n)=\int_0^{\infty} x^{n-1}e^{-x}dx \qquad \text{also} \qquad \int_0^{\infty} x^{n-1}e^{-ax}dx=\frac{\Gamma(n)}{a^n}$$

PROPERTIES OF GAMMA FUNCTION

1. $\Gamma(n)=(n-1).\Gamma(n-1)$
2. $\Gamma(n)=(n-1)!$ *If* 'n' *is a positive* int*eger.*
3. $\Gamma(1)=1$
4. $\Gamma(\frac{1}{2})=\sqrt{\pi}$
5. $\Gamma(-\frac{1}{2})=\frac{1}{\sqrt{\pi}}$

EXAMPLES OF GAMMA FUNCTION

$$\int_0^{\infty} x\ e^{-x}\,dx=\Gamma(2)=1!=1$$

$$\int_0^{\infty} x^2\ e^{-x}\,dx=\Gamma(3)=2!=2$$

$$\int_0^{\infty} x^3\ e^{-2x}\,dx=\frac{\Gamma(4)}{2^4}=\frac{3!}{16}=\frac{6}{16}=\frac{3}{8}$$

$$\int_0^{\infty} x^5\ e^{-3x}\,dx=\frac{\Gamma(6)}{3^6}=\frac{5!}{729}=\frac{120}{729}$$

BETA FUNCTION

$$B(\alpha,\beta)=\int_0^1 x^{\alpha-1}(1-x)^{\beta-1}dx=\frac{\Gamma(\alpha).\Gamma(\beta)}{\Gamma(\alpha+\beta)}$$

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PROPERTIES OF GAMMA DISTRIBUTION

PROPERT-1 AREA UNDER THE GAMMA CURVE

We have to show that $\int_0^\infty f(x)dx = 1$

$$L.H.S = \int_0^\infty f(x)dx$$

$$= \int_0^\infty \frac{\lambda^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\lambda x}\, dx$$

$$= \frac{\lambda^\alpha}{\Gamma(\alpha)} \int_0^\infty x^{\alpha-1} e^{-\lambda x}\, dx$$

$$= \frac{\lambda^\alpha}{\Gamma(\alpha)} \times \frac{\Gamma(\alpha)}{\lambda^\alpha} = 1$$

PROPERT-2 MEAN OF GAMMA DISTRIBUTION

$$Mean = E(X) = \int_0^\infty x\, f(x)dx$$

$$= \int_0^\infty x \frac{\lambda^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\lambda x}\, dx$$

$$= \frac{\lambda^\alpha}{\Gamma(\alpha)} \int_0^\infty x^{1+\alpha-1} e^{-\lambda x}\, dx$$

$$= \frac{\lambda^\alpha}{\Gamma(\alpha)} \int_0^\infty x^{\alpha} e^{-\lambda x}\, dx$$

$$= \frac{\lambda^\alpha}{\Gamma(\alpha)} \times \frac{\Gamma(\alpha+1)}{\lambda^{\alpha+1}}$$

$$= \frac{\lambda^\alpha}{\Gamma(\alpha)} \times \frac{\alpha\,\Gamma(\alpha)}{\lambda^\alpha \lambda^1}$$

$$= \frac{\alpha}{\lambda}$$

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PROPERT-3 VARIANCE OF GAMMA DISTRIBUTION

$$V(X) = E\left(X^2\right) - \left(E\left(X\right)\right)^2 \text{ -------(1)}$$

we have $E\left(X\right) = \alpha/\lambda$

Now $E\left(X^2\right) = \int_0^\infty x^2 \; f\left(x\right) dx$

$$= \int_0^\infty x^2 \frac{\lambda^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\lambda x} \, dx$$

$$= \frac{\lambda^\alpha}{\Gamma(\alpha)} \int_0^\infty x^2 \, x^{\alpha-1} e^{-\lambda x} \, dx$$

$$= \frac{\lambda^\alpha}{\Gamma(\alpha)} \int_0^\infty x^{2+\alpha-1} e^{-\lambda x} \, dx$$

$$= \frac{\lambda^\alpha}{\Gamma(\alpha)} \int_0^\infty x^{\alpha+1} e^{-\lambda x} \, dx$$

$$= \frac{\lambda^\alpha}{\Gamma(\alpha)} \times \frac{\Gamma(\alpha+2)}{\lambda^{\alpha+2}}$$

$$= \frac{\lambda^\alpha}{\Gamma(\alpha)} \times \frac{(\alpha+1)\,\Gamma(\alpha+1)}{\lambda^\alpha \times \lambda^2}$$

$$= \frac{\lambda^\alpha}{\Gamma(\alpha)} \times \frac{(\alpha+1)\,\alpha\,\Gamma(\alpha)}{\lambda^\alpha \times \lambda^2}$$

$$= \frac{\alpha(\alpha+1)}{\lambda^2}$$

Put the values of $E\left(X\right)$ *and* $E\left(X^2\right)$ *in equation* (1), *we get*

$$V(X) = \frac{\alpha(\alpha+1)}{\lambda^2} - \left(\frac{\alpha}{\lambda}\right)^2$$

$$= \frac{\alpha^2+\alpha}{\lambda^2} - \frac{\alpha^2}{\lambda^2}$$

$$= \frac{\alpha^2+\alpha-\alpha^2}{\lambda^2} = \frac{\alpha}{\lambda^2}$$

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PROPERT-4 MOMENT GENERATING FUNCTION

Moment Generating Function is given by

$$M_X(t) = E\left(e^{tx}\right)$$

$$= \int_0^\infty e^{tx} f(x)dx$$

$$= \int_0^\infty e^{tx} \frac{\lambda^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\lambda x} dx$$

$$= \frac{\lambda^\alpha}{\Gamma(\alpha)} \int_0^\infty e^{tx-\lambda x} x^{\alpha-1} dx = \frac{\lambda^\alpha}{\Gamma(\alpha)} \int_0^\infty x^{\alpha-1} e^{-(\lambda-t)x} dx$$

$$= \frac{\lambda^\alpha}{\Gamma(\alpha)} \times \frac{\Gamma(\alpha)}{(\lambda-t)^\alpha} = \frac{\lambda^\alpha}{(\lambda-t)^\alpha} = \left(\frac{\lambda}{\lambda-t}\right)^\alpha$$

Mean using Moment Generating Function is given by

$$E(X) = \left[\frac{d}{dt} M_X(t)\right]_{t=0}$$

$$= \left[\frac{d}{dt}\left(\frac{\lambda}{\lambda-t}\right)^\alpha\right]_{t=0}$$

$$= \left[\lambda^\alpha(-\alpha)(\lambda-t)^{-\alpha-1}(-1)\right]_{t=0} = \left[\lambda^\alpha \alpha(\lambda-t)^{-\alpha-1}\right]_{t=0} = \lambda^\alpha \alpha \times \frac{1}{(\lambda)^{\alpha+1}} = \frac{\alpha}{\lambda}$$

Variance using Moment Generating Function is given by

For variance we shall first find $E\left(X^2\right)$

$$E\left(X^2\right) = \left[\frac{d^2}{dt^2} M_X(t)\right]_{t=0}$$

$$= \left[\frac{d}{dt}\left(\lambda^\alpha \alpha(\lambda-t)^{-\alpha-1}\right)\right]_{t=0}$$

$$= \left[\lambda^\alpha \alpha(-\alpha-1)(\lambda-t)^{-\alpha-2}(-1)\right]_{t=0} = \frac{\alpha(\alpha+1)}{\lambda^2}$$

$$V(X) = E\left(X^2\right) - \left(E(X)\right)^2$$

$$= \frac{\alpha(\alpha+1)}{\lambda^2} - \left(\frac{\alpha}{\lambda}\right)^2 = \frac{\alpha^2+\alpha}{\lambda^2} - \frac{\alpha^2}{\lambda^2} = \frac{\alpha^2+\alpha-\alpha^2}{\lambda^2} = \frac{\alpha}{\lambda^2}$$

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PROPERT-5 CUMULANT GENERATING FUNCTION

Cumulant Generating Function is given by

$$K_X(t) = \ln(M_X(t))$$
$$= \ln\left(\frac{\lambda}{\lambda - t}\right)^{\alpha}$$
$$= \alpha\left[\ln\lambda - \ln(\lambda - t)\right]$$

Mean using Cumulant Generating Function is given by

$$E(X) = \left[\frac{d}{dt}K_X(t)\right]_{t=0}$$
$$= \left[\frac{d}{dt}\alpha(\ln\lambda - \ln(\lambda - t))\right]_{t=0}$$
$$= \alpha\left[0 - \frac{1}{\lambda - t}\times -1\right]_{t=0}$$
$$= \left[\frac{\alpha}{\lambda - t}\right]_{t=0}$$
$$= \frac{\alpha}{\lambda}$$

Variance using Cumulant Generating Function is given by

$$V(X) = \left[\frac{d^2}{dt^2}K_X(t)\right]_{t=0}$$
$$= \left[\frac{d}{dt}\left(\frac{\alpha}{\lambda - t}\right)\right]_{t=0}$$
$$= \left[(-\alpha)(\lambda - t)^{-2}(-1)\right]_{t=0}$$
$$= \left[\alpha(\lambda - t)^{-2}\right]_{t=0}$$
$$= \frac{\alpha}{\lambda^2}$$

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